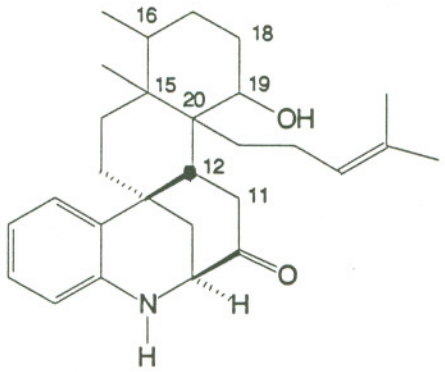
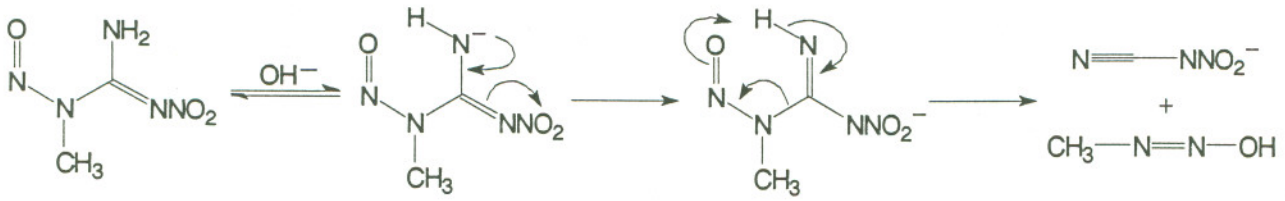
Organic Chemistry Cumulative Exam
March 7, 1992

1. Aspernomine is a newly discovered metabolite from *Aspergillus nomius* with antiinsectan activity (JACS **1992**, *114*, 1015). Its structure is shown below.



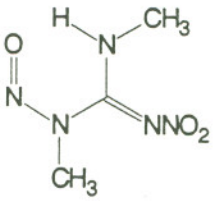
You will note that I did not indicate the stereochemistry at 15, 16, 19, and 20.
NOESY correlation was observed between H-12 and H-16; between H-12 and the axial proton of H-18; and between H-19 and both protons of H-11. (There are also a whole lot of others, but don't worry about them.) Determine the stereochemistry at 15, 16, 19, and 20. Show your reasoning.

2. N-methyl-N'-nitro-N-nitrosoguanidine (MNNG) is a powerful carcinogen which decomposes in basic solution. A possible mechanism for its action is shown below.



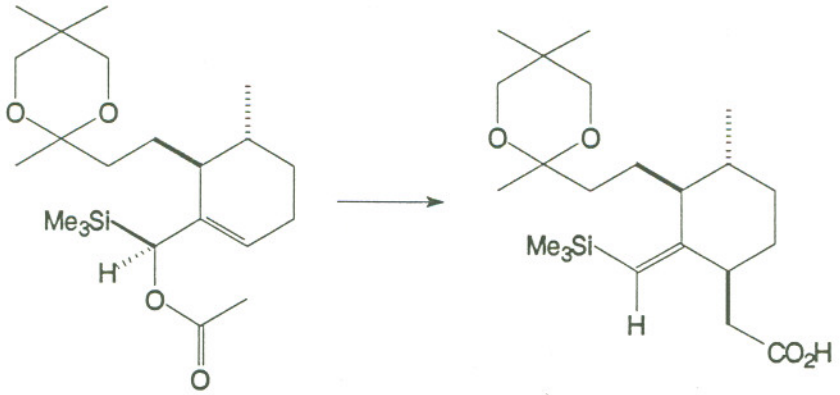
Recently (JACS **1992**, *114*, 1406), kinetic studies of this decomposition process were reported. The first order rate constant for the decomposition of MNNG (k) is very small ($< 10^{-5}$) and pH-independent below pH 6. Above pH 8, log(k) increases linearly with increasing pH with a slope of 1. The pK_a of MNNG is 7.73.

Is this result consistent with the mechanism above? If not, propose an alternative mechanism which *is* consistent. As additional information, the investigators also reported that N,N''-dimethyl-N'-nitro-N-nitrosoguanidine (DMNNG) is substantially more resistant to decomposition under these conditions.



DMNNG

3. The following is one step in the total synthesis of (+)-artemisinin, an antimalarial agent (JACS **1992**, *114*, 974). Using the usual "electron-pushing" diagrams, show the mechanism.



Conditions: diethylamine, n-butyllithium, THF, −78 °C to 23 °C.